1. INTRODUCTION

The present project was carried out to develop and evaluate an Artificial Intelligence–based system for automatic segmentation, bone measurement, and implant size recommendation from dental radiographic images for implant planning. The system aims to extract clinically relevant bone parameters such as bone height, bone width, and cortical bone area using deep learning–based instance segmentation techniques, and to use these measurements for preliminary implant dimension suggestions.

The dataset consisted of CBCT scan images of patients who underwent radiographic examination for dental implant evaluation. The scans were visualized and processed using CS 3D imaging software, and relevant cross-sectional slices were converted into two-dimensional images for further analysis and input into the AI system.

Data Source and Image Acquisition

CBCT images used in this project were obtained from dental radiographic examinations and visualized using CS 3D imaging software. Cross-sectional slices of the mandible representing potential implant sites were selected based on clear visualization of bone structures.

Only images with good contrast, clearly visible mandibular anatomy, and identifiable cortical bone boundaries were considered suitable for analysis. The selected sagittal cross-sectional slices were exported as standard image files (JPG/PNG) and used as input for the AI-based system.

Dataset Preparation

The extracted CBCT slices were organized and reviewed before being used for deep learning analysis. Each image was paired with corresponding segmentation annotations representing key anatomical structures. These images and annotations were formatted for compatibility with the deep learning framework used in this study.

Annotation Procedure

Manual annotation was performed to generate ground truth segmentation masks. Polygon-based masks were drawn around the following anatomical regions:

- Cortical bone
- Mandibular (inferior alveolar) nerve canal

Annotations were carefully traced along anatomical boundaries to ensure accurate representation of bone and nerve structures for model training.

Model Architecture and Training

A Mask R-CNN model with a ResNet-50 Feature Pyramid Network (FPN) backbone was used for automatic instance segmentation. This architecture allows simultaneous detection and pixel-level segmentation of multiple anatomical structures within a single image.

Pretrained weights were used to initialize the model, and it was trained to identify three classes: cortical bone, cancellous bone, and nerve canal. The training process optimized detection and segmentation performance to achieve accurate anatomical delineation.

Inference and Segmentation

After training, the model was applied to new CBCT cross-sectional images. For each image, the system generated:

- Segmentation masks for each detected anatomical structure
- Class labels corresponding to cortical bone, cancellous bone, and nerve canal

The predicted masks were overlaid on the original images to visually confirm correct anatomical segmentation.

Measurement of Jaw Bone Dimensions

Only segmentation masks corresponding to **cortical bone** were used for dimensional measurements, as cortical bone is the primary structure supporting dental implants.

Selection of Cortical Bone Region

In cases where multiple cortical bone regions were detected in a single image, each region was treated as an independent potential implant site. Measurements were performed separately for each cortical bone section to evaluate site-specific bone availability.

For each section, the system calculated:

- **Bone height**
- **Bone width**

These measurements formed the basis for subsequent implant size estimation and recommendation.

Bone Height Estimation (PCA-Based Approach)

Instead of simply selecting the topmost and bottommost pixels in image coordinates, bone height was estimated using a principal component analysis (PCA)–based geometric method, ensuring anatomical orientation invariance.

Step 1: Polygon Extraction

The outer boundary of the cortical bone mask was extracted using contour detection. The largest contour was converted into a polygon representation consisting of ordered boundary coordinates.

Step 2: Principal Axis Determination

The polygon points were centered and a covariance matrix was computed. Eigenvalue decomposition of this matrix was performed:

- The eigenvector corresponding to the largest eigenvalue represents the dominant anatomical axis of the bone section.

- This principal axis corresponds to the natural vertical dimension of the bone, regardless of image tilt.

Step 3: Projection and Height Calculation

All polygon points were projected onto the principal axis. The minimum and maximum projections along this axis were identified to determine:

- Superior boundary (crest)
- Inferior boundary (basal cortical limit)

Bone height in pixels was computed as:

$$\text{Height(px)} = \|\text{Bottom} - \text{Top}\|$$

where the distance is calculated using Euclidean norm:

$$\text{Height(px)} = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$$

This PCA-based approach ensures robust height estimation even in rotated or obliquely oriented bone sections.

Bone Width

Bone Width Measurement

Bone width was measured at a clinically relevant position slightly below the crest, representing the region typically used for implant diameter selection.

Step 1: Offset Position Selection

Starting from the superior boundary (crest), a fixed offset distance (2 mm converted to pixels) was applied along the principal height axis.

Step 2: Perpendicular Sampling

At this offset point, a line perpendicular to the principal height axis was constructed. This perpendicular vector was used to sample points across the mask.

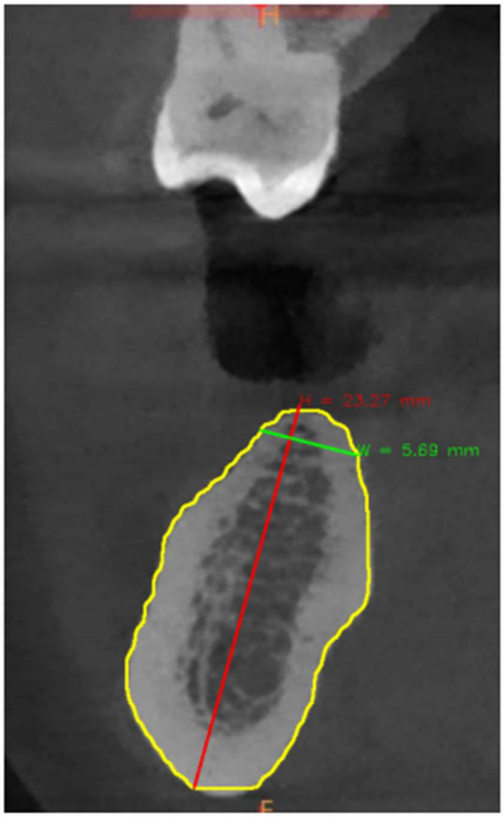
Step 3: Boundary Intersection Detection

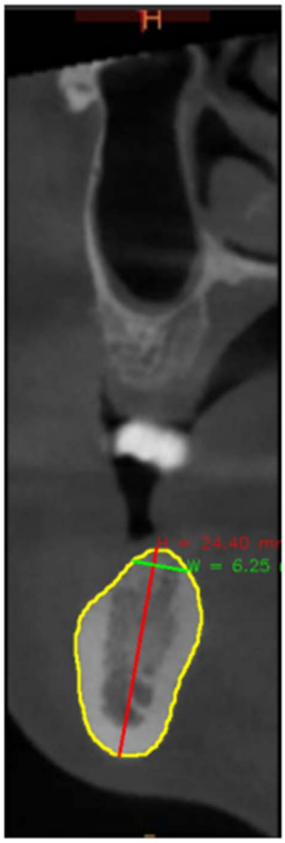
All intersection points between the perpendicular sampling line and the cortical bone mask were collected. The first and last valid intersection points represented the medial and lateral cortical boundaries.

Bone width was calculated as:

$$\text{Width(px)} = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$$

This method ensures that width measurement is geometrically consistent with the anatomical orientation of the bone.





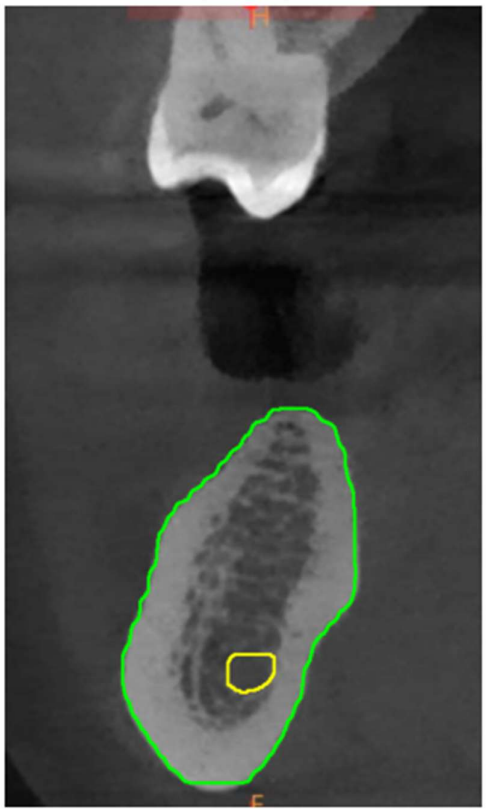
Nerve Canal Segmentation (YOLO-Based Model)

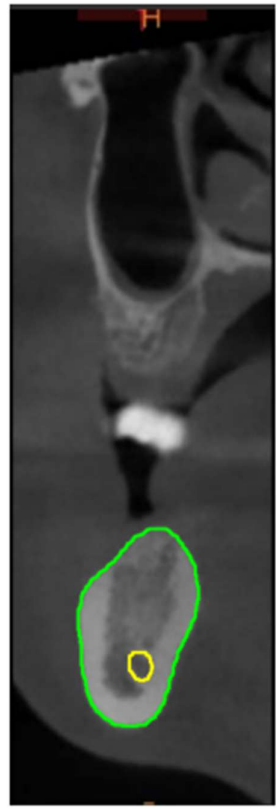
A separate YOLO-based segmentation model was trained specifically for mandibular nerve canal detection.

To generate reliable training labels without manual annotation:

- The color-annotated nerve canal images were processed using saturation-based thresholding.
- Connected component analysis was applied.
- Only the largest anatomically consistent component was retained per slice.

The trained model outputs a binary segmentation mask representing the nerve canal region, enabling automated safety margin estimation during implant planning.





SCALE CALIBRATION USING CS3D ZOOM FACTOR

Since images were obtained from CS3D software with variable zoom levels, a fixed pixel-to-millimeter conversion factor could not be applied.

A known real-world reference distance of approximately 1 mm was used within CS3D to determine pixel density at the selected zoom level. Additionsally, standard screen pixel size was considered as:

$$1\ pixel \approx 0.2645833\ mm$$

Let Zrepresent the zoom factor applied in CS3D during image export.

Then the effective millimeter-per-pixel value is:

$$mm\ per\ pixel = Z \times 0.2645833$$

Final physical measurements were obtained as:

$$Measurement_{mm} = Measurement_{pixels} \times (Z \times 0.2645833)$$

MODEL PERFORMANCE EVALUATION

The performance of the trained Mask R-CNN model was evaluated using standard instance segmentation metrics during training. The model's learning progress was monitored through loss values and segmentation accuracy measures obtained from the training logs.

The total training loss showed a consistent decrease over iterations, indicating effective learning of anatomical structures. Mask segmentation accuracy improved steadily and reached values above **95%** in later training stages, while false positive and false negative rates reduced significantly. These trends demonstrate that the model successfully learned to differentiate cortical bone, cancellous bone, and nerve canal regions.

The final trained model produced smooth and anatomically consistent segmentation masks on validation images, confirming its suitability for automated jaw bone measurement and implant planning support.

Nerve Canal Segmentation Performance

A dedicated YOLOv26 instance segmentation model was trained for mandibular nerve canal detection. Performance was evaluated on the validation dataset using standard segmentation metrics.

The nerve canal segmentation model achieved:

- mAP@50: 76.0%
- Precision: 83.1%
- Recall: 68.2%

The mAP@50 score of 76.0% indicates reliable overlap between predicted masks and ground truth annotations at a 0.5 IoU threshold, which is appropriate for thin, elongated anatomical structures such as the mandibular canal.

The precision of 83.1% demonstrates strong suppression of false positive detections, which is clinically important to avoid overestimation of nerve boundaries during implant planning.

The recall value of 68.2% reflects moderate under-segmentation in certain slices, likely due to variations in contrast, partial visibility, and anatomical complexity of the canal. However, detected regions were geometrically consistent and adequate for safe boundary localization.

Implant Dimension Estimation and Recommendation

After measuring bone height and width, the system estimates a preliminary implant size for each potential implant site.

Implant Length Estimation

In clinical implantology, a minimum safety distance of 2 mm from the mandibular nerve canal is maintained to prevent neurovascular injury.

In the proposed system, implant length is determined using direct geometric analysis of the segmented nerve canal rather than a proportional heuristic.

Computational Procedure

1. The principal bone axis was computed using a PCA-based method from the cortical bone contour.
2. The superior cortical boundary (crest) was identified as the starting point.
3. All segmented nerve canal pixels were projected onto the same principal axis.
4. The nearest nerve projection along this axis was identified.
5. A 2 mm safety offset was applied along the principal axis direction.

The implant length was therefore defined as:

Implant Length = Distance from crest to (nerve position − 2 mm)

Implant Diameter Estimation

Implant diameter was derived from the measured bone width.

Bone width was calculated at a clinically relevant level slightly below the superior cortical boundary using a line perpendicular to the principal bone axis.

To maintain adequate lateral bone support, a 3 mm safety margin (1.5mm from adjacent teeth) was applied: Implant Diameter = Measured Bone Width − 3 mm

This ensures:

- Preservation of cortical bone on both sides
- Reduced risk of buccal or lingual plate perforation
- Improved primary stability conditions

The implant width was aligned perpendicular to the principal bone axis, maintaining geometric consistency with the measured section.

Virtual Implant Placement

A virtual implant representation was generated for visualization using the following geometric rules:

- The implant axis followed the principal bone direction obtained from PCA.
- The implant shoulder originated at the superior cortical boundary.
- The implant tip terminated 2 mm superior to the nerve canal.
- The implant diameter was centered within the measured cortical width.

This resulted in an anatomically aligned virtual implant simulation that respects both vertical and lateral safety margins.

Implant System Recommendation

The computed implant length and diameter were converted from pixels to millimeters using the calibrated pixel-to-millimeter scaling factor derived from CS3D zoom settings.

The estimated dimensions were compared against a structured database of commercially available implant systems, including:

Straumann, Nobel Biocare, Noris, Osstem

For each candidate implant, the system verified:

Diameter ≤ available implant diameter, Length ≤ available implant length

Among all valid options, the implant with the smallest dimensional difference from the computed values was selected as the recommended implant.

The system outputs: Company name, Implant diameter, Implant length

This recommendation serves as a computational decision-support tool and does not replace clinical judgment, radiographic evaluation, or surgical planning expertise.

RESULTS :

	Model : YOLO26 + MASK R-CNN	Manual
Corticol	Height : 23.27 mm Width : 6.3 mm	Height :23.6 mm Width : 5.69 mm
	Height : 23.7 mm Width : 8.7 mm	Height : 24.52 mm Width : 8.15 mm
	Height : 23.3 mm Width : 7.1 mm	Height : 24.40 mm Width : 6.25 mm